1. $b^q \equiv 1 \bmod (b-1)$ applied for every non-negative integer because following the definition of congruence, the above formula implies that $(b-1)$ divides $(b^q - 1)$:
   $(b-1) \mid (b^q - 1)$
   Because the dividend is to the exponent q, every or any non-negative integer, then it will simply just be a multiple of b minus 1 ($b_q \times b_{q-1} \times b_{q-2} \times \ldots b_0 - 1$). This multiple of $b-1$ will always be divisible by $(b-1)$.
   ∴ for every non-negative integer q, we have: $b^q \equiv 1 \bmod (b-1)$

2. Proving: $x \equiv s_x \bmod (b-1)$
   Because $s_x = x_m + x_{m-1} + \cdots + x_1 + x_0$
   And $x = x_m b^m + x_{m-1} b^{m-1} + \cdots + x1\ b + x_0$, $x_m \neq 0$
   That means that :
   $x_m b^m + x_{m-1} b^{m-1} + \cdots + x1\ b + x_0 = x_m + x_{m-1} + \cdots + x_1 + x_0 \bmod (b-1)$
   Following the definition of congruence, the above formula implies that $(b-1)$ divides $(x_m b^m + x_{m-1} b^{m-1} + \cdots + x1\ b + x_0 - x_m + x_{m-1} + \cdots + x_1 + x_0)$
   $(b-1) \mid (x_m b^m + x_{m-1} b^{m-1} + \cdots + x1\ b + x_0 - x_m + x_{m-1} + \cdots + x_1 + x_0)$
   Because the values calculating x and s_x have the summation of the x values, subtracting one form the other will leave the summation of the b values ($b^m + b^{m-1} + \cdots + b$) which would be divisible by $(b-1)$

3. $(z = xy) \rightarrow (s_z \equiv s_x s_y \bmod (b-1))$

   ∴ due to the explanation for the $x \equiv s_x \bmod (b-1)$ proof above, and using the equation that $z = xy$, it can be deduced that multiplying the summations of x (s_x) and y (s_y) and subtracting this value form the summation of z (s_z) would make the value divisible by (b-1).

   Because the summations s_z, s_x, and s_y would have the same values for z, x,and y, simply without the summation of b to the exponent decreasing m value, then dividing the (b-1) from $s_z - s_x s_y$ would leave the same values as $z = xy$.

4. $b = 8$, $x = (2322)_8$, $y = (55640)_8$ and $z = (156324510)_8$

$x = 2(8^3) + 3(8^2) + 2(8^1) + 2(8^0) = 1234_{10}$

$y = 5(8^4) + 5(8^3) + 6(8^2) + 4(8^1) + 0(8^0) = 23456_{10}$

$z = 1(8^8) + 5(8^7) + 6(8^6) + 3(8^5) + 2(8^4) + 4(8^3) + 5(8^2) + 1(8^1) + 0(8^0) = 28,944,712_{10}$

this fits that $z = xy$ because $28, 944, 712 = (1234)(23455)$

∴ $z = xy$ holds